

Mengdi Wang

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Research Summary

My research develops multiscale representations, interface and transport algorithms, and adaptive GPU solvers for physical simulation, while exploring simulation as a two-way interface between AI and the real world: AI expands what can be simulated, and physics provides testable principles for reliable, human-centered intelligent systems.

Research Areas: Physical Simulation, Computer Graphics, Scientific Computing, GPU Computing, Scientific Machine Learning

Academic Appointments

University of British Columbia, Postdoctoral Fellow

2026.9 - Present

• Department of Computer Science, PhysAI Lab, working with Prof. Peter Yichen Chen

Education

Georgia Institute of Technology, Ph.D. in Computer Science

2024.1 - 2026.5

• School of Interactive Computing, advised by Prof. Bo Zhu

Dartmouth College, Ph.D. Program in Computer Science (Transferred)

2020.9 - 2023.12

• Guarini School of Graduate and Advanced Studies, advised by Prof. Bo Zhu

Peking University, B.S. in Computer Science, *summa cum laude*

2016.9 - 2020.6

• Center on Frontiers of Computing Studies, advised by Prof. Baoquan Chen

Publications

WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer

SIGGRAPH Asia 2026

Caoliwen Wang*, Minghao Guo*, Siyuan Chen*, Heng Zhang, *Mengdi Wang*, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, Peter Yichen Chen (*equal contribution)

WorldContact: A Contact-Centric World Model for Scalable Robot Learning

Submitted to ICRA 2027

Caoliwen Wang*, *Mengdi Wang** (co-corresponding), Heng Zhang, Shixun Huang, Siyuan Chen, Chao Liu, Anpei Chen, Zhendong Wang, Peter Yichen Chen (co-corresponding), Huamin Wang (*equal contribution)

WorldAgent: Verification-Guided Agentic Physical World Construction

Submitted to ICLR 2027

Caoliwen Wang*, *Mengdi Wang** (sole corresponding author), Yige Chen*, Zejia Wu, Bowen Huang, Siyuan Chen, Guanxiong Chen, Lifu Wei, Heng Zhang, Qinghai Zhang, Yin Yang, Guandao Yang, Shiyong Xiong, Peng Wang, Chenfanfu Jiang, Peter Yichen Chen (*equal contribution)

Matrix-Free GPU Implementation of Algebraic Multigrid Poisson Solver on Adaptive Octree with Irregular Domains

Submitted to JCP 2026

Mengdi Wang, Yuchen Sun, Bo Zhu

Cirrus: Adaptive Hybrid Particle-Grid Flow Maps on GPU

SIGGRAPH 2025

Mengdi Wang, Fan Feng, Junlin Li, Bo Zhu (Featured in video trailer)

Real-Time Knit Deformation and Rendering

SIGGRAPH 2025

*Mengdi Wang**, Tao Huang*, Haoyang Shi* (joint first authors), Yuxing Qiu, Yin Yang, Kui Wu

An Interface Tracking Method with Triangle Edge Cuts

JCP 2025

Mengdi Wang, Matthew Cong, Bo Zhu

Hydrophobic and Hydrophilic Solid-Fluid Interaction

SIGGRAPH Asia 2022

Jinyuan Liu, *Mengdi Wang*, Fan Feng, Annie Tang, Qiqin Le, Bo Zhu

A Moving Eulerian-Lagrangian Particle Method for Thin Film and Foam Simulation Yitong Deng, <i>Mengdi Wang</i> , Xiangxin Kong, Shiyong Xiong, Zangyueyang Xian, Bo Zhu	SIGGRAPH 2022
A Clebsch Method for Free-Surface Vortical Flow Simulation Shiyong Xiong, Zhecheng Wang, <i>Mengdi Wang</i> , Bo Zhu (Featured in video trailer)	SIGGRAPH 2022
Thin-Film Smoothed Particle Hydrodynamics Fluid <i>Mengdi Wang</i> , Yitong Deng, Xiangxin Kong, Aditya H. Prasad, Shiyong Xiong, Bo Zhu (Featured in video trailer)	SIGGRAPH 2021
Visual Data Analysis and Simulation Prediction for COVID-19 Baoquan Chen, Mingyi Shi, Xingyu Ni, Liangwang Ruan, Hongda Jiang, Heyuan Yao, <i>Mengdi Wang</i> , Zhenhua Song, Qiang Zhou, Tong Ge	arXiv preprint (2020)

Research Experience

Research Intern , Lightspeed Studio, Tencent IEG • Designed a real-time yarn-level knit fabric simulation framework, resulting in a co-first-author SIGGRAPH 2025 paper.	2024.5 - 2024.8
Research Intern , NVIDIA • Designed a GPU-accelerated geometric multigrid Poisson solver and VOF-based interface tracking method, published in <i>Journal of Computational Physics</i> . Mentored by Ken Museth and Eftychios Sifakis.	2022.6 - 2022.9

Teaching Experience

Instructor , Dartmouth College, COSC 294 (Graduate Seminar)	Fall 2021
Additional Instructor , Dartmouth College / Georgia Tech	Multiple terms
Head Teaching Assistant , Dartmouth College, COSC 277 Computer Graphics	Fall 2023

Academic Service

Reviewer: ACM Transactions on Graphics (SIGGRAPH), Computer Graphics Forum (CGF), Journal of Computational Physics

Awards and Honors

ACM-ICPC Asia Qingdao Regional Contest , Gold Medal (Rank 4th)	2016
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